Roll No. | | | | | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem-3)

PHARMACEUTICAL ENGINEERING

Subject Code : BP-304T

M.Code : 93326

Date of Examination : 19-06-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

   a. What are the applications of Bernoulli's theorem?

   b. Write demerits of elutriation tank.

   c. What do you mean by heat exchangers?

   d. Define specific heat and mention its units.

   e. Mention filter aids.

   f. What is Rittingers Rule?

   g. Differentiate between conduction and convection.

   h. Mention the types of glass used for packaging liquids.

   i. Enumerate the causes of corrosion.

   j. Enumerate the factors that influence rate of filtration.

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SECTION-B

2. Enumerate the factors influencing energy loss during flow of fluids. Explain the remedies for them.

3. What is Lyophilization? Discuss the factors influencing this process and mention the applications of this process.

4. Explain the mechanisms of size separation. Write a note on Bag Filters and their applications.

SECTION-C

5. Discuss the operation of a rotary drum filter.

6. What is a climbing film evaporator? Describe its operation and advantages.

7. What is a twin shell blender? Describe its operation and mention its applications.

8. Explain the mechanisms of heat transfer. Discuss the operation of heat exchangers.

9. Illustrate a fluidized bed dryer and explain its operation. Mention its applications in pharmaceutical industry.

10. Discuss the operation of a rotary drum filter.

11. Write a note on non-perforated basket centrifuge and its applications.

12. Explain the mechanisms of heat transfer. Discuss the operation of heat exchangers.

13. Comment on nonferrous materials used in pharmaceutical plants.

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NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

June-2024